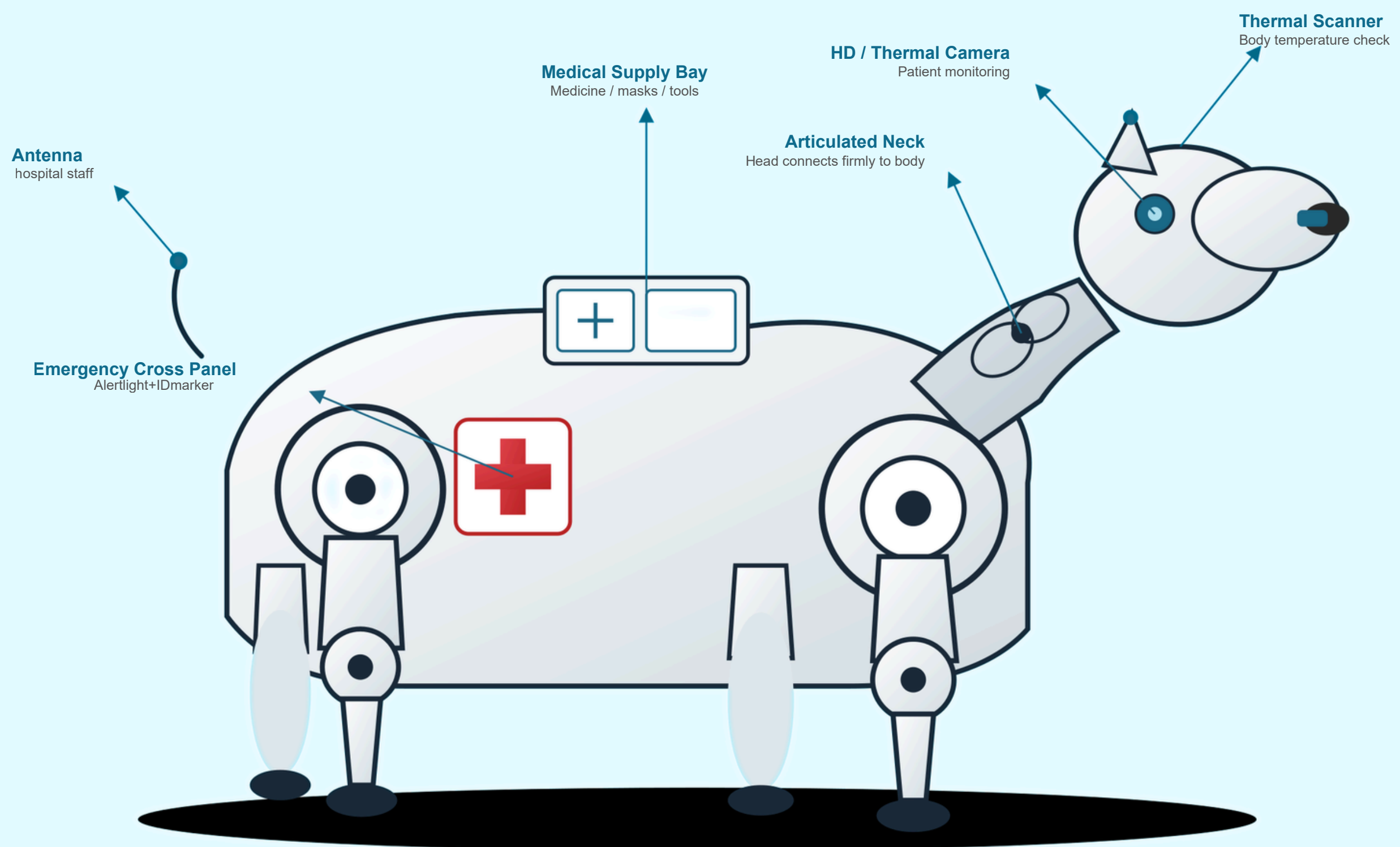
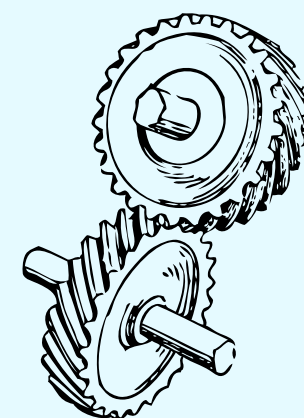


MEDICAL ASSISTANCE ROBOT DOG



Number of Joints and Degrees of Freedom
Each leg contains 2 joints so 2 DOF per leg.
 $4 \text{ legs} \times 2 = 8 \text{ DOF}$



Type of Motor:

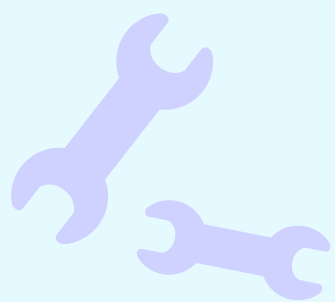
Servo Motor because it provides precise angular control and sufficient torque.

Preliminary Torque Calculation (5 kg robot)

Max load per leg (Trot Gait): 2.5 kg

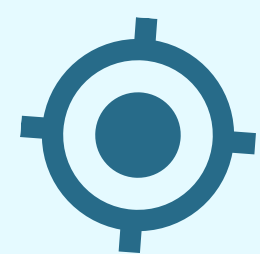
Moment arm: $L \approx 0.12 \text{ m}$

$$t = (2.5 * 9.81) * 0.12 = 2.94, \text{ N.m}$$



Stability and Center of Gravity

Positioned near the center of the body and as low as possible.



Proposed Gait Method

Trot Gait: The front-right leg moves together with the back-left leg, then vice versa.

Expected Mechanical Issues

Servo overload during long standing periods (overheating/stall).
Mechanical backlash accumulation in the joints with repeated use.